

KAVINRAJ S

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Summary

Machine Learning Engineer and final-year Computer Science undergraduate experienced in building production-grade ML systems and full-stack applications end to end. Currently fine-tuning deep learning models at IGCAR using metaheuristic algorithms (Grey Wolf Optimizer, Genetic Algorithms). Comfortable owning the full product lifecycle from MVP to scalable deployment across ML pipelines, distributed systems, and modern web stacks. Seeking to join SprintFour Technologies to help early-stage startups ship robust, AI-powered products quickly.

Education

Amrita Vishwa Vidyapeetham, Coimbatore

June 2023 – May 2027

Bachelor of Technology in Computer Science | CGPA: 9.0/10 | Minor: Internet of Things

Technical Skills

Languages & Query: Python, SQL, Java, C++, Go, JavaScript, TypeScript

AI/ML & Data Science: PyTorch, TensorFlow, scikit-learn, Metaheuristic Optimisation (Grey Wolf, Genetic Algorithms), Hyperparameter Optimisation, Federated Learning (FLwr), Pandas, NumPy, OpenCV, SMOTE

Web & Product: React/Svelte, Node.js, REST APIs, Streamlit, MVP & full-stack delivery, Experiment Automation

Cloud & DevOps: GCP (Cloud Functions, BigQuery, Cloud Storage), CI/CD, Git, Linux, Agile/Scrum

Databases: PostgreSQL, Redis, Neon DB, NoSQL

Experience

IGCAR (Indira Gandhi Centre for Atomic Research)

2025 – Present

Machine Learning Engineer

Kalpakkam, Tamil Nadu, India

- Fine-tuned deep learning models using metaheuristic hyperparameter optimisation, implementing Grey Wolf Optimizer (GWO) and Genetic Algorithms (GA) to search high-dimensional configuration spaces, improving model accuracy by [X]% over grid- and random-search baselines.
- Engineered an automated, population-based tuning pipeline that converged on optimal model configurations in [X]× fewer evaluations than manual search, cutting experiment turnaround and making model optimisation reproducible.

IEEE

Dec 2024 – Mar 2025

Machine Learning Research Intern

Remote

- Conducted a comprehensive study on Federated Learning, analysing privacy-utility tradeoffs across 8 distributed model configurations; quantified accuracy-privacy Pareto frontiers and prepared findings for peer review.
- Automated the end-to-end experiment pipeline, cutting configuration-to-result time from 4 hours to 15 minutes and enabling sweeps across 50+ hyperparameter combinations (3× faster iteration).

XCMG

May 2024 – July 2024

Software Engineering Intern

Tamil Nadu, India

- Developed a Python-based real-time ETL pipeline integrating operational data streams from multiple sources into a unified analytics layer, reducing anomaly detection latency by 75% and eliminating 12+ hours per week of manual data processing across 3 departments.

Projects

Privacy-Preserving Federated Learning System | Python, PyTorch, FLwr, GCP

2024–2025

- Built a distributed ML system training across 10+ simulated clients at 96.8% accuracy (within 2% of the centralised baseline) while enforcing strict local data privacy — a scalable, privacy-compliant architecture for data-sensitive products.
- Implemented FedAvg aggregation over 50K+ distributed training samples with gradient compression, reducing inter-node communication overhead by 40%; deployed and benchmarked on GCP for bandwidth-constrained cloud-edge scenarios.

Polyverse: Federated Decentralised Network | Svelte, Redis, PostgreSQL (Neon DB), DIDs

2025

- Designed a federated platform implementing W3C Decentralised Identifiers (DIDs) for self-sovereign identity management, enabling cross-node interoperability without a centralised authority; applied eventual consistency and distributed data governance.
- Built a persistent storage layer on serverless PostgreSQL with a Redis caching tier, applying data modelling across relational and in-memory NoSQL stores to reduce high-frequency query latency.

AI-Powered Automated Testing Framework | Python, TypeScript, LLMs, VS Code API

2025

- Built a VS Code extension automating unit-test generation for Python and TypeScript, achieving 85%+ code coverage and cutting manual test-writing effort by 60% across 20+ projects.
- Designed a context-aware analysis engine generating 200+ test cases per session at 92% validity, with automated coverage reporting aligned to CI/CD practices.